Daehyun Choi

Postdoctoral Researcher · Fluid Mechanics · Bio-inspired Propulsion · Physical AI

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"Translating biological locomotion into physical intelligence through fluid physics and machine learning"

Summary

I am a postdoctoral researcher at **Georgia Tech**, applying fluid mechanics to vision AI, robotics, digital twins, bioinspiration, and animal behavior. My current research focuses on data-driven shape optimization of physical AI and soft matter.

Highlights

- **Journals** (first-authored): *J. Fluid Mech.* ($\times$3), *Phys. Fluids* (Editor's Pick), *Sci. Rep.*, *Integr. Comp. Biol.*; papers under review and in preparation (arXiv preprints available).
- **Awards:** DARPA Young Faculty Award; Sejong International Postdoctoral Fellowship, NRF Korea ($140k), 2023; Domestic Postdoctoral Scholarship, NRF Korea ($50k), 2022.

Research Interests

- Bioinspired propulsion and aquatic locomotion
- Fluid-structure interaction and vortex dynamics
- Deep learning for flow measurement and optimization
- Multiphase flows and interfacial dynamics
- Physical AI and soft robotics

Skills

- **Hardware:** High-speed imaging, 2D/3D PIV, 3D DIC, stereo-camera reconstruction, dynamometry; PCB design & fabrication, circuit design, BLE wireless sensing, microcontroller data acquisition; silicone elastomer fabrication; biological fieldwork.
- **Software:** C++, Python, MATLAB, Java; CNN / optical flow (PyTorch) — DeepBubbleVelocimetry [★ **9**], PWC-Net-based velocity measurement for bubble-laden flows [paper]; FSI simulation (OpenFOAM + CalculiX + preCICE) [arXiv]; Bayesian/multi-objective optimization (BoTorch); multiphysics simulation (ANSYS Fluent, MATLAB).

Education

Seoul National University — Seoul, Korea
M.S. & Ph.D. in Mechanical Engineering — 2022

Seoul National University — Seoul, Korea
B.S. in Mechanical Engineering — 2015

Academic Employment

Georgia Tech — Atlanta, GA, USA
Postdoctoral Researcher — 2023 – present

Ajou University — Suwon, Korea
Postdoctoral Researcher — 2023

Publications

Bold denotes the presenting/corresponding author. † denotes equal contribution. * denotes co-first authorship.

Journal Articles

1. **Daehyun Choi**, Kai Lauren Yung, Halley Wallace, Jacob Harrison, & M. Saad Bhamla, "Interfacial dynamics on the water-hopping behavior of mudskipper," *In preparation.*
2. **Daehyun Choi**, Paras Singh, Halley Wallace, Chandan Bose, & M. Saad Bhamla, "Deep-learning Bayesian surrogate model for thrust enhancement of soft nozzle," *Phys. Rev. Fluids*. *Under review.*
3. **Daehyun Choi***, Paras Singh, Ian Bergerson, Minho Kim, Jieun Park, Halley J. Wallace, Kenny Zhang, Sandy Y. Hsieh, Aqua T. Asberry, Theodore A. Uyeno, William F. Gilly, Hyungmin Park, Daeshik Kang, Chandan Bose, & M. Saad Bhamla, "Squid-inspired soft superpropulsion," *arXiv*:2605.03239 (2026). *Under review.* arXiv
4. Paras Singh, **Daehyun Choi**, M. Saad Bhamla, & Chandan Bose, "Elastic wave propagation governs impulse enhancement in pulsed jets through flexible nozzles," *arXiv*:2605.17319 (2026). *Under review.* arXiv
5. Pankaj Rohilla, Johnathan Nathaniel O'Neil, Chandan Bose, Victor M. Ortega-Jimenez, **Daehyun Choi**, & M. Saad Bhamla, "Epineuston vortex recapture enhances thrust in tiny water skaters," *Under review.* PDF bioRxiv
6. **Daehyun Choi**, Kai Yung, Ian Bergerson, Halley Wallace, Ulmar Grafe, & M. Saad Bhamla, "Three-dimensional tracking method for water-hopping mudskippers in natural habitats," *Integr. Comp. Biol.*, icaf139 (2025). PDF DOI
7. Pankaj Rohilla*, **Daehyun Choi***, Halley Wallace, Kai Lauren Yung, Juhi Deora, Atharva Lele, & M. Saad Bhamla (*equal contribution), "Mastering the Manu – How humans create large splashes," *J. R. Soc. Interface*, 15, 2 (2025). PDF DOI
8. **Daehyun Choi** & Hyungmin Park, "Enhanced impulse and entrainments of the pulsed circular jet by flexible nozzle," *J. Fluid Mech.*, 996, A6 (2024). PDF DOI
9. **Daehyun Choi***, Hyunseok Kim*, & Hyungmin Park (*equal contribution), "Bubble velocimetry using the conventional and CNN-based optical flow algorithms," *Sci. Rep.*, 12, 11879 (2022). PDF DOI Code
10. **Daehyun Choi** & Hyungmin Park, "Flow-structure interaction of the starting jet through the flexible circular nozzle," *J. Fluid Mech.*, 949, A39 (2022). PDF DOI
11. **Daehyun Choi**, Jungwon Byun, & Hyungmin Park, "Analysis of the liquid column atomization by annular dual-nozzle gas jet flow," *J. Fluid Mech.*, 943, A25 (2022). PDF DOI
12. Dokeun Lee, **Daehyun Choi**, Hyungmin Park, & Sungjae Kim, "Electroconvective circulating flows by asymmetric coulombic force distribution in multiscale porous membrane," *J. Membr. Sci.*, 636, 119286 (2021). PDF DOI
13. **Daehyun Choi** & Hyungmin Park, "Flow around in-line sphere array at moderate Reynolds number," *Phys. Fluids* **(Editor's Pick)**, 30, 097104 (2018). PDF DOI

Conferences & Seminars

1. "Hydrodynamics of water-hopping behavior of mudskippers," *2026 Society for Integrative and Comparative Biology (SICB)*, Oral presentation (Symposium). USA, 2026.
2. "Squid-inspired flexible nozzles improve thrust in rotary propulsors using a multi-fidelity approach," *2026 Society for Integrative and Comparative Biology (SICB)*, Oral presentation. USA, 2026.
3. "Squid funnel-inspired passively flexible nozzles for underwater propulsion," *2026 Society for Integrative and Comparative Biology (SICB)*, Oral presentation. USA, 2026.
4. "Geometrical effects of free-end nozzles on wave reflection in self-oscillating flexible nozzles," *78th Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Houston, USA, 2025.
5. "Bio-inspired flexible nozzle design for enhanced underwater propulsion," *Fluid and Biodynamics Seminar, Max Planck Institute for Dynamics and Self-Organization*, Oral presentation (Invited). Göttingen, Germany, 2025.

6. "Squid-inspired nozzles for enhanced thrust in rotary propulsors," *2025 European Mechanics Society: Bio-inspired Fluid-Structure Interaction (EUROMECH)*, Oral presentation. Vienna, Austria, 2025.
7. "Squid-inspired nozzles for enhanced thrust in rotary propulsors," *2025 American Physical Society March Meeting*, Oral presentation. Anaheim, USA, 2025.
8. "Non-invasive 3D tracking of water-hopping mudskippers in natural habitats," *2025 Society for Integrative and Comparative Biology (SICB)*, Poster presentation. Atlanta, USA, 2025.
9. "Physical modelling of water-hopping mudskipper," *77th Annual Meeting of the APS Division of Fluid Dynamics*, Poster presentation. Salt Lake City, USA, 2024.
10. "Water-hopping behavior of mudskipper," *2024 IUTAM Symposium on Capillarity and Elastocapillarity in Biology*, Poster presentation. Seoul, Korea, 2024.
11. "Hydrodynamics of water-hopping mudskipper," *2024 International Physics of Living Systems Annual Meeting*, Oral presentation. Trieste, Italy, 2024.
12. "Hydrodynamic study on the water-hopping behavior of mudskipper," *2024 American Physical Society March Meeting*, Oral presentation. Minneapolis, USA, 2024.
13. "Vortex mechanism in microvelia-inspired water-walking for thrust optimization using robotic design," *2024 Society for Integrative and Comparative Biology (SICB)*, Poster presentation. Seattle, USA, 2024.
14. "Experimental study on the starting jet through the flexible circular nozzle," *2022 World Congress on Advances in Civil, Environmental and Materials Research (ACEM)*, Oral presentation. Seoul, Korea, 2022.
15. "On the optimal flexibility condition of an impulsive starting jet nozzle for thrust generation," *73rd Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Online, 2020.
16. "Experimental study on the effect of nozzle flexibility on the evolution of a starting liquid jet," *72nd Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Seattle, USA, 2019.
17. "Study on dual-nozzle jet flows for the application of liquid atomization," *71st Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Atlanta, USA, 2018.
18. "Experimental study on the flow around in-line array of spheres," *69th Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Portland, USA, 2016.

Research Project Experience

Squid-Inspired Nozzles for Enhanced Efficiency and Thrust in Rotary Propulsors — Georgia Tech
Supported by **Defense Advanced Research Projects Agency** — 2024 – present

Hydrodynamics of Water-Hopping Mudskipper and Biomimetic Robot Development — Georgia Tech
Supported by **National Research Foundation of Korea** – Sejong International Postdoctoral Fellowship — 2023 – present

Development of Cephalopod-Inspired Jet-Propelled Soft Underwater Vehicle — Ajou University
Supported by **National Research Foundation of Korea** – Domestic Postdoctoral Fellowship — 2022 – 2023

Physical-Based Model for Film Thickness and Spray Characteristics Depending on Operation Conditions — Seoul National University
Supported by Daewoo Shipbuilding & Marine Engineering (DSME) — 2021 – 2022

Apparatus for Determining Removal of Foley Catheter — Seoul National University
Supported by Yonsei Severance Hospital — 2020 – 2022

Measurement of Bubble and Particle in Liquid — Seoul National University
Supported by Samsung Electro-Mechanics — 2018 – 2019

Analysis of Molten Metal Atomization Using Multi-physics Simulation — Seoul National University
Supported by Samsung Electronics — 2018 – 2018

Effects of Superhydrophobic Surface on the Wake of Circular Cylinder — Seoul National University
Supported by Samsung Advanced Institute of Technology (SAIT) — 2016 – 2016

Awards & Grants

2025	**Emerging Scholar Best Paper Prize (Top 10),** *Journal of Fluid Mechanics*, **Cambridge University Press**
2024	**Technical Leader — DARPA Young Faculty Award ($500k), Defense Advanced Research Projects Agency**
2024	**Robert M. Nerem International Travel Award ($3k), Georgia Tech**
2024	**Young Researcher Travel Award ($1k),** IUTAM Symposium on Capillarity and Elastocapillarity in Biology
2023	**Sejong International Postdoctoral Fellowship ($140k), National Research Foundation of Korea**
2022	**Best Thesis Award,** Dept. of Mechanical Engineering, **Seoul National University**
2022	**Domestic Postdoctoral Scholarship ($50k), National Research Foundation of Korea**
2016	**BK21+ Doctoral Scholarship ($10k),** Ministry of Education, Korea
2015	**Presidential Scholarship for Undergraduates ($30k),** Korea
2014	**1st Prize — EDISON Challenge (CFD contest, $1k),** Ministry of Science, ICT and Future Planning, Korea

Mentoring & Teaching

Graduate (3) and undergraduate (5) students; 1 high school student — Georgia Tech
Research Mentor — 2024 – present

3 undergraduate students — Seoul National University
Undergraduate Thesis Advisor — 2016 – 2021

Mechanical Engineering Laboratory: Flow Measurements — Seoul National University
Teaching Assistant — 2016 – 2017

Patents (Lead Inventor)

Compliant Nozzle System and Method for Rotary Propulsion — Georgia Tech Research Corporation
US Patent Application, No. 19/568,682 — 2026

Compliant Nozzle System and Method for Fluid Jet Impulse — Georgia Tech Research Corporation
US Patent Application, No. 19/568,695 — 2026

System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with flow and hydraulic pressure — Korea
KR Patent, 10-2021-0149171 — 2021

System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with hydraulic pressure and UASIF — Korea
KR Patent, 10-2021-0149176 — 2021

System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with UASIF — Korea
KR Patent, 10-2021-0149175 — 2021

Apparatus for determining removal of Foley catheter — International
PCT Patent, PCT/KR2020/018848 — 2020

System for reducing ureteral insertion pressure by artificially triggering hydronephrosis based on measurement of pressure associated with the insertion of the ureteral access sheath for RIRS — Korea
KR Granted Patent, 10-2020-0064729 — 2020

System for reducing ureteral entry pressure by generating vibrations in the guide wire for the insertion of the ureteral access sheath for RIRS — Korea
KR Granted Patent, 10-2020-0064731 — 2020

Apparatus for determining removal of Foley catheter — Korea
KR Granted Patent, 10-2020-0000288 — 2020